Exponential Families

Teaching move

Open Lecture 2 with the established course title and subtitle.

Layout sketch

lecture-title

lecture-footer

Content objects

lecture-title — Lecture title and subtitle — card

Lecture 2

Exponential Families

Common coordinates for probability models

lecture-footer — Abbreviated lecture title — card

Exponential Families

Reveal sketch

Static: lecture-title, lecture-footer

Design constraints

Static, restrained course title-slide treatment; no new mathematics. This lecture-level unit is separate from the 53 section slides and eight section design-notes pages.

SECTION DESIGN NOTES

Section 1 — Exponential families give familiar distributions a common form

Purpose

Introduce a useful common coordinate system for familiar distributions.

Endpoint

Students can read $p(x|\eta) = h(x)\exp(\eta^T T(x) - A(\eta))$, identify the four roles, derive the Bernoulli representation, and explain why Bernoulli's natural parameter is log odds.

Entry / assumed context

Students know the Bernoulli model well from prior Bayesian statistics and have just reviewed Gaussian distributions in Lecture 1.

Narrative arc

1. Open with the approved section title and subtitle.
2. Familiar distributions look different but admit a common algebraic form.
3. Name the four recurring roles in that form.
4. Recover those roles by algebraically rewriting Bernoulli.
5. Interpret the Bernoulli natural coordinate and its normalizer.
6. Retell the general-form ⟶ Bernoulli-specialization argument in the summary.

Visual and notation anchors

- Primary anchor: $p(x|\eta) = h(x)\exp(\eta^T T(x) - A(\eta))$.
- Secondary anchor: the Bernoulli correspondence $T(x) = x$, $\eta = \log\left(\frac{\mu}{1-\mu}\right)$, $h(x) = 1$, $A(\eta) = \log(1 + e^{\eta})$.
- Interpretive anchor: $\mu \in (0, 1) \leftrightarrow \eta \in \mathbb{R}$.
- Keep μ for Bernoulli probability and η for the natural parameter.

Summary design brief

Retell **shared template ⟶ Bernoulli mapping ⟶ log odds**. Select the general equation and Bernoulli correspondence as visual anchors, with takeaways for the four roles and natural coordinates. Gaussian algebra is excluded.

Continuity / entry context

Enter from familiar Bernoulli probabilities and the Gaussian density. Retain the general form on the worked Bernoulli slide. End retrospectively; Section 2 owns its opening connection.

Constraints / open pressure

Six slides in 10–12 minutes. Gaussian is motivation only; no derivatives or MLE. Use $0 < \mu < 1$ for finite natural coordinates; endpoints are limits. Source: Murphy Book 2 §§2.4.1–2.4.2.1, printed pp.34–35, equations (2.160) and (2.168)–(2.173).

Section 1 — title

Exponential families give familiar distributions a common form

Teaching move

Open the question of how familiar distributions can be described in common coordinates using the approved title and subtitle.

Layout sketch

section-title

footer-title

Content objects

section-title — Section title and subtitle — card

Exponential families give familiar distributions a common form

Bernoulli reveals the shared roles of data statistics, parameters, and normalization.

footer-title — Abbreviated footer title — card

Exponential-family form.

Reveal sketch

Static: section-title, footer-title

Design constraints

The full title is the dominant title-page object; the subtitle is subordinate directly beneath it. The abbreviated title specifies section footer metadata, not a second body panel. Use the established course section-title treatment. Do not add mathematics, an agenda, or a key-point region to this opening.

Different distributions can share the same algebraic structure

Teaching move

Motivate exponential-family coordinates by placing two familiar distributions beside a single shared algebraic skeleton.

Layout sketch

bernoulli | gaussian

shared

takeaway

Content objects

bernoulli — Bernoulli — card

$$\mathrm{Ber}(x \mid \mu) = \mu^x (1-\mu)^{1-x}, \quad x \in \{0, 1\}$$

gaussian — Gaussian — card

$$\mathcal{N}(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

shared — Shared exponential-family form — card

$$p(x \mid \eta) = h(x) \exp(\eta^T T(x) - A(\eta))$$

takeaway — Key points — key-points

- Familiar distributions can look very different in their usual parameterizations.
- Exponential-family form rewrites them using one common algebraic structure.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	bernoulli, gaussian	—	Initial meaningful context
STATE 2	+ shared	entry emphasis	Expose the common algebraic target.
STATE 3	+ takeaway	—	State the common-structure claim.

Summary candidates

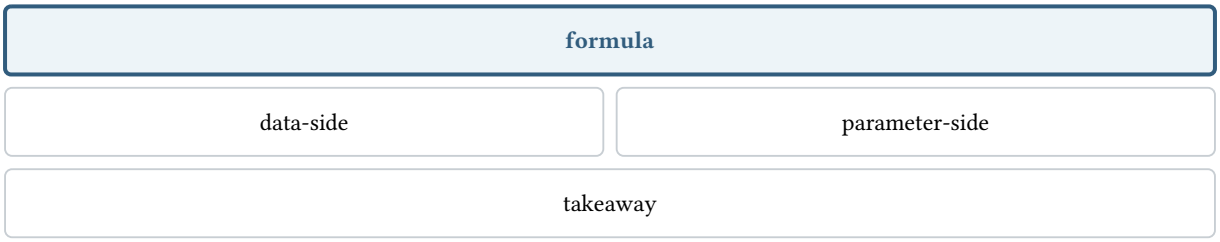
VISUAL shared — The general exponential-family equation is the section's main visual anchor.
KEY POINT (takeaway) Exponential-family form gives familiar distributions one common algebraic structure.

Exponential families separate data statistics from natural parameters

Teaching move

State the general form, then organize its four named roles into the data-side and parameter-side pieces asserted by the title.

Layout sketch



Content objects

formula — Exponential-family form — card

$$p(x \mid \eta) = h(x)\exp(\eta^T T(x) - A(\eta))$$

$$A(\eta) = \log Z(\eta)$$

data-side — Data-side functions — bordered-card

$h(x)$ — **base measure**: fixed with respect to η
$T(x)$ — **sufficient statistics**: the data representation

parameter-side — Parameter-side terms — bordered-card

η — **natural parameters**: pair linearly with $T(x)$
$A(\eta)$ — **log-partition function**: normalizes the distribution

takeaway — Key points — key-points

- Data-side $h(x), T(x)$ meet parameter-side $\eta, A(\eta)$ through the linear pairing $\eta^T T(x)$.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	formula	—	Initial meaningful context
STATE 2	+ data-side	entry emphasis	Identify the two data-side roles together.
STATE 3	+ parameter-side	entry emphasis	Identify the two parameter-side roles together.
STATE 4	+ takeaway	—	State the separation and the linear interaction.

Summary candidates

KEY POINT (takeaway) Data-side $h(x), T(x)$ meet parameter-side $\eta, A(\eta)$ through $\eta^T T(x)$.

Bernoulli already has the exponential-family form

Teaching move

Work from the ordinary Bernoulli probability to the minimal exponential-family representation so that $T(x) = x$ and the log-odds natural parameter emerge from familiar algebra.

Layout sketch

- form-recall
- bernoulli-start
- exponential-rewrite
- **collect-x**
- mapping
- takeaway

Content objects

form-recall — Carried forward

recall-card

$$p(x \mid \eta) = h(x)\exp(\eta^T T(x) - A(\eta))$$

From: Previous slide: exponential-family form

bernoulli-start — Familiar Bernoulli model

card

$$\mathrm{Ber}(x \mid \mu) = \mu^x (1-\mu)^{1-x}, \quad x \in \{0,1\}, \quad 0 < \mu < 1$$

exponential-rewrite — Rewrite powers as an exponential

card

$$= \exp(x \log \mu + (1-x)\log(1-\mu))$$

collect-x — Collect the coefficient of x

card

$$= \exp\left(x \log\left(\frac{\mu}{1-\mu}\right) + \log(1-\mu)\right)$$

mapping — Match terms to the general form

bordered-card

$$T(x) = x, \quad \eta = \log\left(\frac{\mu}{1-\mu}\right), \quad h(x) = 1$$

$$A(\eta) = -\log(1-\mu) = \log(1+e^{\eta})$$

takeaway — Key points

key-points

Rewriting the same Bernoulli probability exposes $T(x) = x$ and $\eta = \log\left(\frac{\mu}{1-\mu}\right)$; the probability model itself is unchanged.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	form-recall, bernoulli-start	—	Initial meaningful context
STATE 2	+ exponential-rewrite	entry emphasis	Expose the exponent of the familiar probability.
STATE 3	+ collect-x	entry emphasis	Separate the data term from the parameter-only term.
STATE 4	+ mapping	entry emphasis	Match the collected expression to the persistent general form.
STATE 5	+ takeaway	—	State the meaning of the rewrite.

Summary candidates

VISUAL mapping — The Bernoulli correspondence is the section's clearest worked specialization.
KEY POINT (takeaway) Rewriting Bernoulli exposes $T(x) = x$ and $\eta = \log\left(\frac{\mu}{1-\mu}\right)$ without changing the model.

Design constraints

The three labeled algebra steps form one aligned derivation in production, with earlier lines retained. Keep the general-form recall subordinate. Reveal mapping only after the coefficient of x is visible; transient entry emphasis settles on subsequent states. Do not put borders around individual algebra lines. The final state must show the whole derivation and mapping.

Bernoulli log odds are its natural parameter

Teaching move

Interpret the carried-forward natural parameter as log odds, invert the coordinate change, and identify its normalizer.

Layout sketch

eta-recall

coordinates | normalizer

takeaway

Content objects

eta-recall — Carried forward recall-card

$$\eta = \log\left(\frac{\mu}{1-\mu}\right)$$

From: Previous slide: Bernoulli exponential-family mapping

coordinates schema-card

$$\eta = \operatorname{logit}(\mu) \quad \leftrightarrow \quad \mu = \sigma(\eta) = \frac{1}{1+e^{-\eta}}$$

$$\mu \in (0,1) \quad \leftrightarrow \quad \eta \in \mathbb{R}$$

The natural parameter is an unconstrained coordinate for the same Bernoulli family.

normalizer — Normalization in natural coordinates bordered-card

$$Z(\eta) = \sum_{x=0}^{1} e^{x\eta} = 1 + e^{\eta}$$

$$A(\eta) = \log(1+e^{\eta}), \quad p(x \mid \eta) = \exp(x\eta - A(\eta))$$

takeaway — Key points key-points

- η is the **log odds**: it maps a constrained probability to an unconstrained real coordinate.
- $A(\eta) = \log(1+e^{\eta})$ normalizes Bernoulli in that coordinate system.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	eta-recall, coordinates	—	Initial meaningful context
STATE 2	+ normalizer	entry emphasis	Add the normalization term once the coordinate change is understood.
STATE 3	+ takeaway	—	State the interpretation of the natural coordinate.

Summary candidates

KEY POINT (takeaway) η is log odds, mapping $\mu \in (0,1)$ to $\eta \in \mathbb{R}$.

Section 1 — summary

One representation exposes the same ingredients across families

Teaching move

Retell the section as a general-form ⟶ Bernoulli-specialization ⟶ natural-coordinate argument using only already-established anchors.

Layout sketch

summary-visual

summary-takeaways

Content objects

summary-visual — Summary visual — section-summary-visual

$$p(x \mid \eta) = h(x)\exp(\eta^T T(x) - A(\eta))$$

$\boldsymbol{h(x)}$	1
$\boldsymbol{T(x)}$	x
$\boldsymbol{\eta}$	$\log\left(\frac{\mu}{1-\mu}\right)$
$\boldsymbol{A(\eta)}$	$\log(1+e^{\eta})$

summary-takeaways — Summary takeaways — section-summary-takeaways

- The exponential-family form separates data-side functions from parameter-side coordinates.
- Bernoulli has $h(x) = 1$, $T(x) = x$, $\eta = \log\left(\frac{\mu}{1-\mu}\right)$, and $A(\eta) = \log(1 + e^{\eta})$.
- Its natural parameter is log odds: $\mu \in (0, 1) \leftrightarrow \eta \in \mathbb{R}$.

Reveal sketch

Static: summary-visual, summary-takeaways

Rationale

Select the general formula and the Bernoulli correspondence from the nominated visual anchors. Together they tell the section's general ⟶ specific story directly. Keep the log-odds interpretation in the takeaways rather than adding a third visual, which would weaken the summary's hierarchy.

Design constraints

This is the final Section 1 teaching move. Select only material established on the preceding slides; include no forward pointer or closing handoff. Use the course's explicitly labeled summary treatment while preserving the approved slide title.

SECTION DESIGN NOTES

Section 2 — A Gaussian can be rewritten in exponential-family coordinates

Purpose

Derive every piece of the Gaussian exponential-family representation from the familiar density in one self-contained section.

Endpoint

Students can identify Gaussian $h(\boldsymbol{x})$, $T(\boldsymbol{x})$, η, and $A(\eta)$, and connect $\Lambda = \Sigma^{-1}$ and $\xi = \Lambda\mu$ to the density's linear and quadratic coefficients.

Entry / assumed context

Section 1 supplied the common form. Lecture 1 supplied the multivariate Gaussian density, mean, covariance, and quadratic forms. Introduce precision and information coordinates here; prior exposure to them is not assumed.

Narrative arc

Familiar density ⟶ precision and expanded quadratic ⟶ linear and quadratic data statistics ⟶ natural coefficients and normalizer ⟶ complete mapping.

Visual and notation anchors

- $\boldsymbol{x}, \mu \in \mathbb{R}^D$; Σ and Λ are symmetric positive definite.
- $\Lambda = \Sigma^{-1}$; $\xi = \Lambda\mu$.
- $T(\boldsymbol{x}) = \left[\boldsymbol{x}; \operatorname{vec}(\boldsymbol{x}\boldsymbol{x}^T)\right]$.
- $\eta = \left[\xi; -\frac{1}{2}\operatorname{vec}(\Lambda)\right]$.
- Use $h(\boldsymbol{x}) = (2\pi)^{-\frac{D}{2}}$ throughout.

Summary design brief

Select the complete four-role mapping as the summary visual. Retain the meanings of precision and precision-weighted mean in two short takeaways. Do not repeat all quadratic-expansion lines or introduce moments.

Continuity / entry context

Open by testing Section 1's common form on the known Gaussian density. This unit stands alone if moved to a recording; end retrospectively.

Constraints / open pressure

Six slides in 8–10 minutes. No conditioning, inference, or derivatives of A. Vectorization stacks matrix entries in one fixed order; the full symmetric-matrix representation is nonminimal. Source: Murphy Book 2 §2.4.2.5, printed pp.38–39, equations (2.198)–(2.213).

Precision and information coordinates are defined locally, consistent with the corrected upstream prerequisite contract.

Section 2 — title

A Gaussian can be rewritten in exponential-family coordinates

Teaching move

Frame the Gaussian rewrite as a test of the common form established in Section 1.

Layout sketch

section-title

footer-title

Content objects

section-title — Section title and subtitle — card

A Gaussian can be rewritten in exponential-family coordinates

From the familiar density to precision and information coordinates.

footer-title — Abbreviated footer title — card

Gaussian coordinates.

Reveal sketch

Static: section-title, footer-title

Design constraints

Use the established section-title treatment: full title dominant, subtitle beneath it, abbreviated title as footer metadata. This title page contains no agenda or new mathematical content.

The Gaussian density hides an exponential-family representation

Teaching move

Place the known Gaussian density against the common-form target and name the rewrite students will perform.

Layout sketch

gaussian-density

common-form

takeaway

Content objects

gaussian-density — The familiar multivariate Gaussian

card

$$p(\boldsymbol{x} \mid \mu, \Sigma) = \frac{1}{(2\pi)^{\frac{D}{2}} \, |\Sigma|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(\boldsymbol{x}-\mu)^T \Sigma^{-1} (\boldsymbol{x}-\mu)\right)$$

$$\boldsymbol{x}, \mu \in \mathbb{R}^D, \quad \Sigma = \Sigma^T > 0$$

common-form — Carried forward

recall-card

$$p(\boldsymbol{x} \mid \eta) = h(\boldsymbol{x}) \exp(\eta^T T(\boldsymbol{x}) - A(\eta))$$

From: Section 1: common exponential-family form

takeaway — Key points

key-points

The rewrite separates functions of the data from parameter coefficients and normalization.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	gaussian-density	—	Initial meaningful context
STATE 2	+ common-form	—	Make the representation target visible.
STATE 3	+ takeaway	—	Identify the algebraic task.

Design constraints

The complete familiar density is the main inspection object; the recalled target is subordinate. State that $\Sigma > 0$ means positive definite. Preserve the Lecture 1 notation: bold data vector, unbold mean vector.

Expanding the quadratic separates data-dependent and parameter-only terms

Teaching move

Introduce precision, expand the quadratic, and collect the data terms apart from the parameter-only terms of the density.

Layout sketch

density-recall

precision

quadratic-expansion

grouped-density

takeaway

Content objects

density-recall — Carried forward

recall-card

$$p(\boldsymbol{x} \mid \mu, \Sigma) = \frac{1}{(2\pi)^{\frac{D}{2}} \ |\Sigma|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(\boldsymbol{x}-\mu)^T \Sigma^{-1}(\boldsymbol{x}-\mu)\right)$$

From: Previous slide: Gaussian density

precision — Precision replaces inverse covariance

card

$$\Lambda = \Sigma^{-1}, \quad \Lambda = \Lambda^T > 0$$

$$|\Sigma|^{-\frac{1}{2}} = |\Lambda|^{\frac{1}{2}} = \exp\left(\frac{1}{2}\log|\Lambda|\right)$$

quadratic-expansion — Expand, then use symmetry

card

$$\begin{aligned}(\boldsymbol{x}-\mu)^T\Lambda(\boldsymbol{x}-\mu) &= \boldsymbol{x}^T\Lambda\boldsymbol{x} - \boldsymbol{x}^T\Lambda\mu - \mu^T\Lambda\boldsymbol{x} + \mu^T\Lambda\mu \\ &= \boldsymbol{x}^T\Lambda\boldsymbol{x} - 2\boldsymbol{x}^T\Lambda\mu + \mu^T\Lambda\mu\end{aligned}$$

grouped-density — The expanded log density

schema-card

$$\begin{aligned}\log p(\boldsymbol{x} \mid \mu, \Sigma) = &-\frac{D}{2}\log(2\pi) \\ &+\underbrace{\boldsymbol{x}^T\Lambda\mu - \frac{1}{2}\boldsymbol{x}^T\Lambda\boldsymbol{x}}_{\text{data-dependent}} \\ &+\underbrace{\frac{1}{2}\log|\Lambda| - \frac{1}{2}\mu^T\Lambda\mu}_{\text{parameter-only}}\end{aligned}$$

takeaway — Key points key-points

The data enter through a linear term and a quadratic term; the remaining terms depend only on the parameters.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	density-recall	—	Initial meaningful context
STATE 2	+ precision	entry emphasis	Define precision and rewrite the determinant factor.
STATE 3	+ quadratic-expansion	entry emphasis	Use symmetry to combine the two cross terms.
STATE 4	+ grouped-density	entry emphasis	Collect the three roles in the log density.
STATE 5	+ takeaway	—	State what expansion exposed.

Summary candidates

VISUAL grouped-density — Shows the origin of data terms and normalization.

Design constraints

Retain the original density as compact subordinate context. Treat each expansion or grouping as one teaching state, keeping both expansion lines visible together. Underbrace labels are explanatory annotations. Do not make the determinant identity a separate derivation.

The Gaussian sufficient statistics are x and xx^T

Teaching move

Rewrite the quadratic term as a coefficient–statistic pairing, then identify the linear and quadratic sufficient statistics.

Layout sketch

data-terms

quadratic-pairing

statistics

takeaway

Content objects

data-terms — Carried forward recall-card

$$\boldsymbol{x}^T \Lambda \mu - \frac{1}{2} \boldsymbol{x}^T \Lambda \boldsymbol{x}$$

From: Previous slide: data-dependent exponent

quadratic-pairing — Expose the quadratic data object card

$$\boldsymbol{x}^T \Lambda \boldsymbol{x} = \sum_{i,j} \Lambda_{ij} x_i x_j = \operatorname{vec}(\Lambda)^T \operatorname{vec}(\boldsymbol{x}\boldsymbol{x}^T)$$

$$\left(\boldsymbol{x}\boldsymbol{x}^T\right)_{ij} = x_i x_j$$

statistics — One vector of linear and quadratic statistics schema-card

$$T(\boldsymbol{x}) = \begin{bmatrix} \boldsymbol{x} \\ \operatorname{vec}(\boldsymbol{x}\boldsymbol{x}^T) \end{bmatrix}$$

$\operatorname{vec}(M)$: stack the columns of M

takeaway — Key points key-points

Gaussian sufficient statistics contain the coordinates and their pairwise products: $\boldsymbol{x}$ and $\boldsymbol{x}\boldsymbol{x}^T$.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	data-terms	—	Initial meaningful context
STATE 2	+ quadratic-pairing	entry emphasis	Expose the entries of the quadratic statistic.
STATE 3	+ statistics	—	Stack the two data objects into the common-form statistic vector.
STATE 4	+ takeaway	—	Name what the statistics retain.

Summary candidates

VISUAL statistics — Supplies the Gaussian data side of the final mapping.

KEY POINT (takeaway) Gaussian sufficient statistics are the coordinates and their pairwise products.

Design constraints

Column stacking is a bookkeeping convention. Display the equality of the quadratic form, entry sum, and vector pairing simultaneously. Mention once that symmetric matrices repeat off-diagonal entries, so this full-vector representation is nonminimal; do not derive a minimal parameterization.

Precision and precision-weighted mean are Gaussian natural parameters

Teaching move

Match the coefficients to the statistic vector, then convert the parameter-only exponent into the log-partition term.

Layout sketch

statistics-recall

natural-parameters | normalization

canonical-density

takeaway

Content objects

statistics-recall — Carried forward recall-card

$$T(\boldsymbol{x}) = \begin{bmatrix} \boldsymbol{x} \\ \operatorname{vec}(\boldsymbol{x}\boldsymbol{x}^T) \end{bmatrix}$$

From: Previous slide: Gaussian sufficient statistics

natural-parameters — Natural coordinates schema-card

$$\xi = \Lambda\mu, \quad \eta = \begin{bmatrix} \xi \\ -\frac{1}{2}\operatorname{vec}(\Lambda) \end{bmatrix}$$

$$\eta^T T(\boldsymbol{x}) = \boldsymbol{x}^T\xi - \frac{1}{2}\boldsymbol{x}^T\Lambda\boldsymbol{x}$$

$$\Sigma = \Lambda^{-1}, \quad \mu = \Lambda^{-1}\xi$$

normalization — Parameter-only terms give −A bordered-card

$$\mu^T\Lambda\mu = \xi^T\Lambda^{-1}\xi$$

$$-A(\eta) = \frac{1}{2}\log|\Lambda| - \frac{1}{2}\xi^T\Lambda^{-1}\xi$$

$$h(\boldsymbol{x}) = (2\pi)^{-\frac{D}{2}}$$

canonical-density — Complete exponential-family form card

$$p(\boldsymbol{x} \mid \eta) = h(\boldsymbol{x})\exp(\eta^T T(\boldsymbol{x}) - A(\eta))$$

$$A(\eta) = -\frac{1}{2}\log|\Lambda| + \frac{1}{2}\xi^T\Lambda^{-1}\xi$$

takeaway — Key points key-points

- Precision Λ controls the quadratic coefficient; precision-weighted mean ξ controls the linear coefficient.
- (ξ, Λ) are information coordinates for the same Gaussian.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	statistics-recall	—	Initial meaningful context
STATE 2	+ natural-parameters	entry emphasis	Identify the coefficients paired with the statistics.
STATE 3	+ normalization	entry emphasis	Read the base measure and negative log partition from the expanded density.
STATE 4	+ canonical-density	—	Assemble all four common-form roles.
STATE 5	+ takeaway	—	Interpret the Gaussian natural coordinates.

Summary candidates

VISUAL natural-parameters — Pairs the data statistics with their Gaussian coefficients.
VISUAL canonical-density — Completes the Gaussian common-form mapping.
KEY POINT (takeaway) Precision controls the quadratic coefficient; precision-weighted mean controls the linear coefficient.

Design constraints

Keep the statistic vector subordinate while the natural coefficients become the dominant object. Then compare coefficients and normalization side by side. With $h(\boldsymbol{x}) = (2\pi)^{-\frac{D}{2}}$, A has no additional $\frac{D}{2}\log(2\pi)$ term. Require Λ symmetric positive definite. Introduce the name “information coordinates” here without claiming it was taught in Lecture 1.

Section 2 — summary

Gaussian information coordinates are exponential-family coordinates

Teaching move

Retell the completed rewrite as a single four-role mapping from ordinary Gaussian parameters to exponential-family coordinates.

Layout sketch

summary-visual

summary-takeaways

Content objects

summary-visual — Summary visual

section-summary-visual

$$\Lambda = \Sigma^{-1}, \quad \xi = \Lambda\mu$$

$$p(\boldsymbol{x}|\eta) = h(\boldsymbol{x})\exp(\eta^T T(\boldsymbol{x}) - A(\eta))$$

$\boldsymbol{h(x)}$	$(2\pi)^{-\frac{D}{2}}$
$\boldsymbol{T(x)}$	$\begin{bmatrix} \boldsymbol{x} \\ \mathrm{vec}(\boldsymbol{x}\boldsymbol{x}^T) \end{bmatrix}$
$\boldsymbol{\eta}$	$\begin{bmatrix} \xi \\ -\frac{1}{2}\,\mathrm{vec}(\Lambda) \end{bmatrix}$
$\boldsymbol{A(\eta)}$	$-\frac{1}{2}\log\lvert\Lambda\rvert + \frac{1}{2}\xi^T\Lambda^{-1}\xi$

summary-takeaways — Summary takeaways

section-summary-takeaways

- Gaussian sufficient statistics are the coordinates and their pairwise products.
- Precision controls the quadratic coefficient; precision-weighted mean controls the linear coefficient.

Reveal sketch

Static: summary-visual, summary-takeaways

Rationale

Select the statistic vector, natural coefficients, and completed canonical density. Recombine them into one mapping; omit the nominated expansion because it would repeat the algebra rather than summarize its result.

Design constraints

This final teaching move is retrospective. Preserve the same base-measure convention and full-vector representation used above. Include the course's explicit summary label without changing the approved title. No moments, derivative identities, or closing handoff appear here.

SECTION DESIGN NOTES

Section 3 — The log-partition function connects parameters to moments

Purpose

Derive the mean identity from normalization, then interpret natural and mean coordinates and sketch the covariance identity.

Endpoint

Students can derive $\nabla A(\eta) = E_\eta[T(X)]$, read mean parameters, and interpret $\nabla^2 A(\eta) = \mathrm{Cov}_\eta[T(X)]$ as covariance and convexity.

Entry / assumed context

Use the general form and Bernoulli mapping from Section 1. Gaussian algebra is optional context; the argument does not depend on Section 2 being live.

Narrative arc

Normalization for each parameter → differentiate under the integral → mean statistic → Bernoulli coordinate check → covariance and convexity → summary.

Visual and notation anchors

- $p(x|\eta) = h(x)\exp(\eta^T T(x) - A(\eta))$.
- $m(\eta) = E_\eta[T(X)] = \nabla A(\eta)$.
- $\nabla^2 A(\eta) = \mathrm{Cov}_\eta[T(X)]$.
- Bernoulli: $T(x) = x$, $A(\eta) = \log(1 + e^\eta)$, $m = \mu$.

Summary design brief

Select the paired mean/covariance identities as the dominant visual. Retain Bernoulli's logistic map as a compact check, with a convexity takeaway. Omit the full differentiation sequence from the summary.

Continuity / entry context

Open by asking what Section 1's normalizer must do when the natural weights change. Close by consolidating only the model-moment identities taught here.

Constraints / open pressure

Seven slides in 18–20 minutes. Fixed parameter-independent support, interior natural parameters, finite required moments, and valid differentiation under the integral are assumed. Replace integrals by sums for discrete models. State covariance/convexity without claiming strict convexity. Do not claim a unique general inverse unless the family is minimal; Bernoulli's inverse is explicit. No conjugate functions, Fisher information, or MLE. Source: Murphy Book 2 §§2.4.3–2.4.4, printed pp.39–41, equations (2.216)–(2.230) and (2.233)–(2.237).

Section 3 — title

The log-partition function connects parameters to moments

Teaching move

Frame the question of what normalization reveals about model statistics.

Layout sketch

section-title

footer-title

Content objects

section-title — Section title and subtitle — card

The log-partition function connects parameters to moments

Normalization links natural parameters to means and covariance.

footer-title — Abbreviated footer title — card

Log-partition moments.

Reveal sketch

Static: section-title, footer-title

Design constraints

Use the shared section-title hierarchy, with the abbreviated title as footer metadata. No new formula or key-point panel appears on this title page.

The normalizer must change when the natural parameter changes

Teaching move

Expose the normalization identity that holds at every valid natural parameter, providing the fixed identity to differentiate.

Layout sketch

- form-recall
- partition
- **normalization**
- takeaway

Content objects

form-recall — Carried forward — recall-card

$$p(x|\eta) = h(x)\exp(\eta^T T(x) - A(\eta))$$

From: Section 1: common form

partition — Normalization determines A — card

$$Z(\eta) = \int h(x)\exp(\eta^T T(x))\,\mathrm{d}x, \quad A(\eta) = \log Z(\eta)$$

normalization — One identity for every valid parameter — card

$$\int p(x|\eta)\,\mathrm{d}x = \exp(-A(\eta))Z(\eta) = 1$$

For discrete models, replace the integral by a sum.

takeaway — Key points — key-points

$A(\eta)$ compensates for the natural weights so total probability stays one.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	form-recall	—	Initial meaningful context
STATE 2	+ partition	—	Identify the unnormalized mass and its logarithm.
STATE 3	+ normalization	entry emphasis	Make the invariant total probability explicit.
STATE 4	+ takeaway	—	State the role of the normalizer.

Summary candidates

VISUAL normalization — The fixed identity starts the derivative argument.

Design constraints

The support is fixed with respect to η; all displayed identities apply where $Z(\eta) < \infty$. The normalization identity becomes the next slide's subordinate recall. Do not assert a nonzero change in every parameter direction.

Differentiating normalization exposes the expected sufficient statistic

Teaching move

Differentiate the constant total probability, then use the common-form derivative to expose an expectation of the statistic.

Layout sketch

normalization-recall

differentiate

inside-derivative

expectation

takeaway

Content objects

normalization-recall — Carried forward recall-card

$$\int p(x|\eta)\,\mathrm{d}x = 1$$

From: Previous slide: total probability

differentiate — Differentiate both sides card

$$0 = \nabla_\eta \int p(x|\eta)\,\mathrm{d}x = \int \nabla_\eta p(x|\eta)\,\mathrm{d}x$$

inside-derivative — Derivative of the common-form density card

$$\nabla_\eta p(x|\eta) = p(x|\eta)(T(x) - \nabla A(\eta))$$

expectation — Recognize the expectation card

$$\begin{aligned} 0 &= \int (T(x) - \nabla A(\eta)) p(x|\eta)\,\mathrm{d}x \\ &= E_\eta[T(X)] - \nabla A(\eta) \end{aligned}$$

takeaway — Key points key-points

Differentiating normalization equates $\nabla A(\eta)$ with the expected statistic.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	normalization-recall	—	Initial meaningful context
STATE 2	+ differentiate	entry emphasis	The derivative of total probability is zero.
STATE 3	+ inside-derivative	entry emphasis	The chain rule produces the statistic minus the normalizer gradient.
STATE 4	+ expectation	entry emphasis	Integrate the density-weighted statistic and use normalization.
STATE 5	+ takeaway	—	Name the equality the calculation established.

Design constraints

Preserve the normalization identity and prior derivative steps throughout. The final two lines form one aligned equation group. State the regularity assumption when moving the derivative inside the integral; do not expand it into a separate theorem. For discrete models the same steps use sums.

The gradient of the log-partition function is the mean parameter

Teaching move

Separate the reusable mean identity from the calculation and interpret the mapping from natural weights to expected statistics.

Layout sketch

mean-identity

coordinate-map

takeaway

Content objects

mean-identity — Mean parameters — card

$$m(\eta) = E_\eta[T(X)] = \nabla A(\eta)$$

coordinate-map — From natural weights to model means — schema-card

$$\eta \overset{\nabla A}{\to} m$$

Natural parameters weight the statistics; mean parameters state their expectations.

takeaway — Key points — key-points

The mean parameter is the expected sufficient statistic, which need not be $E[X]$.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	mean-identity	—	Initial meaningful context
STATE 2	+ coordinate-map	—	Interpret the direction of the parameter map.
STATE 3	+ takeaway	—	Distinguish a statistic mean from the raw-data mean.

Summary candidates

VISUAL mean-identity — The section's first reusable derivative identity.
KEY POINT (takeaway) The mean parameter is the expected sufficient statistic.

Design constraints

Show a one-way map here; do not assert a unique inverse for a redundant representation. The general mean parameter may be a vector.

Bernoulli shows natural and mean parameters are two coordinates for one family

Teaching move

Differentiate the known Bernoulli log partition and identify its result as both the mean statistic and the inverse log-odds map.

Layout sketch

bernoulli-recall

derivative

coordinate-map

takeaway

Content objects

bernoulli-recall — Carried forward recall-card

$$T(x) = x, \quad A(\eta) = \log(1 + e^{\eta}), \quad \eta = \log\left(\frac{\mu}{1-\mu}\right)$$

From: Section 1: Bernoulli mapping

derivative — Recover the model mean card

$$A'(\eta) = \frac{e^{\eta}}{1+e^{\eta}} = \frac{1}{1+e^{-\eta}} = \mu = E_{\eta}[X]$$

coordinate-map — Two coordinates, one Bernoulli model schema-card

$$\eta \in \mathbb{R} \overset{\text{logistic}}{\rightarrow} m = \mu \in (0,1)$$

$$\mu \in (0,1) \overset{\text{log odds}}{\rightarrow} \eta = \log\left(\frac{\mu}{1-\mu}\right)$$

takeaway — Key points key-points

Bernoulli's natural coordinate is log odds; its mean coordinate is the success probability.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	bernoulli-recall	—	Initial meaningful context
STATE 2	+ derivative	entry emphasis	Verify the mean identity in the known example.
STATE 3	+ coordinate-map	—	Interpret the explicit reversible map.
STATE 4	+ takeaway	—	State the meaning of the two coordinates.

Summary candidates

VISUAL derivative — A compact concrete check of the mean identity.

Design constraints

Keep μ as the Bernoulli probability, matching Section 1. Finite natural coordinates correspond to $0 < \mu < 1$. The visible inverse belongs to this minimal example, not to every possible exponential-family representation.

Differentiating again turns curvature into covariance

Teaching move

Sketch the second derivative using the density derivative, recognize the covariance, and state its positive-semidefinite consequence.

Layout sketch

mean-recall

covariance-sketch

curvature

takeaway

Content objects

mean-recall — Carried forward recall-card

$$m = E_\eta[T(X)] = \nabla A(\eta), \quad \nabla_\eta p = p(T - m)$$

From: Earlier in this section: mean and density derivatives

covariance-sketch — Differentiate the expected statistic card

$$\begin{aligned}\nabla^2 A(\eta) &= \int T(x)(T(x) - m)^T p(x|\eta)\,\mathrm{d}x \\ &= E_\eta[T(X)T(X)^T] - mm^T \\ &= \mathrm{Cov}_\eta[T(X)]\end{aligned}$$

curvature — Covariance gives nonnegative curvature card

$$v^T \nabla^2 A(\eta) v = \mathrm{Var}_\eta[v^T T(X)] \geq 0$$

takeaway — Key points key-points

- Curvature of A is the covariance of the sufficient statistics.
- A is convex; the covariance need not be positive definite.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	mean-recall	—	Initial meaningful context
STATE 2	+ covariance-sketch	entry emphasis	Recognize the second moment minus the outer product of the mean.
STATE 3	+ curvature	—	Read convexity from nonnegative variance in every direction.
STATE 4	+ takeaway	—	Retain covariance and convexity without a strictness claim.

Summary candidates

VISUAL covariance-sketch — The section's second derivative identity.
KEY POINT (takeaway) Covariance gives positive-semidefinite curvature, so A is convex.

Design constraints

This is a derivational sketch, not a second long live proof. Show the three sketch lines together; retain the recall. Do not add a Fisher-information interpretation or claim strict convexity for a nonminimal representation.

Section 3 — summary

The log-partition function converts parameters into moments

Teaching move

Consolidate the paired derivative identities with the Bernoulli mean check.

Layout sketch

summary-visual

summary-takeaways

Content objects

summary-visual — Summary visual — section-summary-visual

$$\nabla A(\eta) = E_\eta[T(X)]$$

$$\nabla^2 A(\eta) = \mathrm{Cov}_\eta[T(X)]$$

Bernoulli: $A'(\eta) = \frac{1}{1+e^{-\eta}} = \mu$.

summary-takeaways — Summary takeaways — section-summary-takeaways

- Normalization makes derivatives of A encode model statistics.
- The gradient gives mean statistics; curvature gives their covariance.
- Covariance makes A convex.

Reveal sketch

Static: summary-visual, summary-takeaways

Rationale

Select the mean identity, covariance result, and Bernoulli derivative. Omit the nominated normalization algebra; the takeaways recall its role without competing with the paired results.

Design constraints

Final retrospective teaching move, explicitly labeled as a summary in production. No dataset statistics, estimation condition, or closing handoff.

SECTION DESIGN NOTES

Section 4 — Learning an exponential-family model means matching model moments to data

Purpose

Derive one general IID maximum-likelihood rule and check it against Bernoulli.

Endpoint

Students can collect $T(\mathcal{D}) = \sum_{n=1}^{N} T(x_n)$, derive the score, and identify the finite-interior-MLE condition $E_{\hat{\eta}}[T(X)] = \frac{T(\mathcal{D})}{N}$.

Entry / assumed context

Section 3 supplied $\nabla A(\eta) = E_\eta[T(X)]$ and convexity of A. Students know IID samples, likelihoods, and the Bernoulli sample proportion.

Narrative arc

IID product ⟶ aggregate statistic and log likelihood ⟶ derivative identity ⟶ score as discrepancy ⟶ moment matching ⟶ Bernoulli check ⟶ summary.

Visual and notation anchors

- $\mathcal{D} = \{x_1, \ldots, x_N\}$, with N fixed.
- $T(\mathcal{D}) = \sum_{n=1}^{N} T(x_n)$; $\overline{T} = \frac{T(\mathcal{D})}{N}$.
- $m(\eta) = E_\eta[T(X)]$.
- $\nabla \ell(\eta) = T(\mathcal{D}) - N m(\eta)$.

Summary design brief

Retell data ⟶ aggregate statistic ⟶ score discrepancy ⟶ fitted mean with one compact equation chain. Keep Bernoulli as a short check rather than another main visual. Preserve the finite-interior qualification.

Continuity / entry context

Open by applying the common form to an observed sample. Close on the established likelihood learning rule; no maximum-entropy handoff.

Constraints / open pressure

Seven slides in 16–18 minutes. Fixed support and regular differentiation assumptions carry forward. Data reduction is a likelihood factorization, not literal equality of dataset and statistic densities. Concavity alone guarantees neither existence nor uniqueness of a finite MLE. Boundary Bernoulli fits have infinite limiting natural coordinates. Source: Murphy Book 2 §2.4.5, printed p.42, equations (2.242)–(2.249); use the factorization and qualified optimum statements described here.

Section 4 — title

Learning an exponential-family model means matching model moments to data

Teaching move

Frame how an observed sample determines parameters within a specified family.

Layout sketch

section-title

footer-title

Content objects

section-title — Section title and subtitle card

Learning an exponential-family model means matching model moments to data

IID likelihoods turn observed statistics into an estimation rule.

footer-title — Abbreviated footer title card

Moment matching.

Reveal sketch

Static: section-title, footer-title

Design constraints

Use the established section-title hierarchy with abbreviated footer metadata. No mathematical argument appears on the title page.

An IID sample enters the likelihood through its aggregate sufficient statistic

Teaching move

Form the IID product, collect repeated terms, and expose the aggregate sufficient statistic in the log likelihood.

Layout sketch

iid-product

collected-product

aggregate

log-likelihood

takeaway

Content objects

iid-product — An IID sample from one model

card

$$\mathcal{D} = \{x_1, ..., x_N\}, \quad p(\mathcal{D}|\eta) = \prod_{n=1}^{N} p(x_n|\eta)$$

$$p(x_n|\eta) = h(x_n)\exp(\eta^T T(x_n) - A(\eta))$$

collected-product — Collect the repeated terms

card

$$p(\mathcal{D}|\eta) = \left(\prod_{n=1}^{N} h(x_n)\right)\exp\left(\eta^T \sum_{n=1}^{N} T(x_n) - NA(\eta)\right)$$

aggregate — Aggregate sufficient statistic

schema-card

$$T(\mathcal{D}) = \sum_{n=1}^{N} T(x_n)$$

log-likelihood — Log likelihood

card

$$\ell(\eta) = \eta^T T(\mathcal{D}) - NA(\eta) + C(\mathcal{D})$$

$$C(\mathcal{D}) = \sum_{n=1}^{N} \log h(x_n)$$

takeaway — Key points

key-points

For fixed N, all parameter-dependent data information enters through $T(\mathcal{D})$.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	iid-product	—	Initial meaningful context
STATE 2	+ collected-product	entry emphasis	Collect repeated statistic and normalization terms.
STATE 3	+ aggregate	—	Name the sample statistic.
STATE 4	+ log-likelihood	entry emphasis	Separate the constant data term from parameter-dependent terms.
STATE 5	+ takeaway	—	State the likelihood reduction.

Summary candidates

VISUAL aggregate — The data side of the learning rule.

Design constraints

The collected product and log likelihood are one continuous derivation. $C(\mathcal{D})$ is independent of η. Preserve the factorization; never replace it with $p(\mathcal{D}|\eta) = p(T(\mathcal{D})|\eta)$. Assume observed data lie in the common support with positive base-measure weight.

Differentiating the likelihood reuses the log-partition moment identity

Teaching move

Differentiate the carried-forward log likelihood and substitute Section 3's mean identity to obtain the score.

Layout sketch

likelihood-recall

mean-recall

differentiate

substitute

takeaway

Content objects

likelihood-recall — Carried forward recall-card

$$\ell(\eta) = \eta^T T(\mathcal{D}) - NA(\eta) + C(\mathcal{D})$$

From: Previous slide: log likelihood

mean-recall — Carried forward recall-card

$$\nabla A(\eta) = E_\eta[T(X)]$$

From: Section 3: mean identity

differentiate — Differentiate with the data fixed card

$$\nabla_\eta \ell(\eta) = T(\mathcal{D}) - N\nabla A(\eta)$$

substitute — Substitute the model moment card

$$\nabla_\eta \ell(\eta) = T(\mathcal{D}) - NE_\eta[T(X)]$$

takeaway — Key points key-points

The derivative of the normalizer supplies the model's expected statistic.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	likelihood-recall, mean-recall	—	Initial meaningful context
STATE 2	+ differentiate	entry emphasis	The data constant drops out; the linear term contributes the aggregate statistic.
STATE 3	+ substitute	entry emphasis	Reuse the already-derived moment identity.
STATE 4	+ takeaway	—	Identify the role of normalization in learning.

Design constraints

Keep both recall equations subordinate and visible while the two score lines form an aligned progression. No new model-specific derivative is needed.

The likelihood gradient compares empirical and model statistics

Teaching move

Interpret the score as a discrepancy by displaying empirical and model statistics simultaneously.

Layout sketch

score

empirical | model

takeaway

Content objects

score — The score measures a discrepancy — card

$$\nabla_\eta \ell(\eta) = T(\mathcal{D}) - N E_\eta[T(X)] = N\big(\overline{T} - m(\eta)\big)$$

empirical — Empirical mean statistic — bordered-card

$$\overline{T} = \frac{1}{N}\sum_{n=1}^{N} T(x_n)$$

Computed from the observed sample.

model — Model mean statistic — bordered-card

$$m(\eta) = E_\eta[T(X)]$$

Expected under the current parameter.

takeaway — Key points — key-points

The score is zero when empirical and model mean statistics agree.

Reveal sketch

Static: score, empirical, model, takeaway

Summary candidates

VISUAL score — The central empirical–model comparison.
KEY POINT (takeaway) The score compares empirical and model mean statistics.

Design constraints

Comparison is simultaneous. Keep the equation dominant and place the two meanings beneath it at equal visual rank. The sample statistic stays fixed while the model mean depends on η.

At the MLE, model moments match empirical moments

Teaching move

Set the score to zero at a finite interior optimum and foreground the resulting general moment-matching condition.

Layout sketch

score-recall

zero-score

matched-moments

concavity-note

takeaway

Content objects

score-recall — Carried forward recall-card

$$\nabla \ell(\eta) = N\big(\overline{T} - m(\eta)\big)$$

From: Previous slide: likelihood gradient

zero-score — If a finite interior MLE exists card

$$\nabla \ell(\hat{\eta}) = 0 \to m(\hat{\eta}) = \overline{T}$$

matched-moments — Moment matching card

$$E_{\hat{\eta}}[T(X)] = \frac{1}{N}\sum_{n=1}^{N} T(x_n)$$

concavity-note — A brief curvature consequence card

$$\nabla^2 \ell(\eta) = -N\,\mathrm{Cov}_{\eta}[T(X)] \leq 0$$

Concavity does not guarantee existence or uniqueness of a finite maximizer.

takeaway — Key points key-points

A finite interior MLE matches expected sufficient statistics to their empirical averages.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	score-recall	—	Initial meaningful context
STATE 2	+ zero-score	entry emphasis	Apply the interior optimum condition.
STATE 3	+ matched-moments	entry emphasis	State the reusable learning rule.
STATE 4	+ concavity-note, takeaway	—	Retain the learning rule with its existence qualification.

Summary candidates

VISUAL matched-moments — The central estimation result.
KEY POINT (takeaway) A finite interior MLE matches model and empirical mean statistics.

Design constraints

Moment matching is the dominant result; keep the curvature consequence as a subordinate note. The matrix inequality means negative semidefinite. Do not turn concavity into another teaching move or an existence theorem.

Bernoulli moment matching recovers the sample proportion

Teaching move

Apply the general rule to Bernoulli's single statistic and recover the familiar probability MLE, including its boundary interpretation.

Layout sketch

bernoulli-recall

proportion

natural-fit

takeaway

Content objects

bernoulli-recall — Carried forward recall-card

$$T(x) = x, \quad E_\eta[T(X)] = \mu$$

From: Sections 1 and 3: Bernoulli statistic and mean

proportion — Observed successes determine the fitted probability card

$$k = \sum_{n=1}^{N} x_n, \quad \hat{\mu} = \frac{k}{N}$$

natural-fit — Natural-coordinate interpretation card

$$0 < k < N: \quad \hat{\eta} = \log\left(\frac{k}{N-k}\right)$$

If $k = 0$ or $k = N$, the probability MLE is 0 or 1; its natural coordinate is a limit $\eta \to -\infty$ or $\eta \to +\infty$.

takeaway — Key points key-points

Bernoulli moment matching is the familiar count-and-divide estimate.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	bernoulli-recall	—	Initial meaningful context
STATE 2	+ proportion	entry emphasis	Match the success probability to the observed fraction.
STATE 3	+ natural-fit	—	Interpret finite and boundary natural coordinates.
STATE 4	+ takeaway	—	Recognize the familiar estimator.

Summary candidates

VISUAL proportion — The compact familiar check of the general rule.

Design constraints

This is a brief specialization, not a new Bernoulli likelihood derivation. Mention boundary cases to interpret the earlier interior qualification.

Section 4 — summary

Maximum likelihood in an exponential family is moment matching

Teaching move

Consolidate aggregate statistics, the score discrepancy, and the qualified moment-matching result as one learning chain.

Layout sketch

summary-visual

summary-takeaways

Content objects

summary-visual — Summary visual

section-summary-visual

$$\mathcal{D} \to T(\mathcal{D}) = \sum_{n=1}^{N} T(x_n)$$

$$\nabla \ell(\eta) = T(\mathcal{D}) - N E_\eta[T(X)]$$

At a finite interior MLE:

$$E_{\hat{\eta}}[T(X)] = \frac{T(\mathcal{D})}{N}$$

Bernoulli check: $\hat{\mu} = \frac{k}{N}$.

summary-takeaways — Summary takeaways

section-summary-takeaways

- The sample enters learning through its aggregate sufficient statistic.
- The score compares observed and model-expected statistics.
- A finite interior MLE matches model and empirical mean statistics.

Reveal sketch

Static: summary-visual, summary-takeaways

Rationale

Select the aggregate statistic, score, and matched-moment result. Compress the Bernoulli nomination to one line; omit the product algebra and concavity note so the full learning chain remains dominant.

Design constraints

Final retrospective teaching move, with the explicit course summary label. Keep the existence qualification visible; no maximum-entropy teaser or notebook handoff belongs here.

SECTION DESIGN NOTES

Section 5 — Moment constraints and maximum entropy lead to exponential families

Purpose

Construct a distribution from expected-feature constraints using minimum KL to a reference, with maximum entropy as the uniform-reference case.

Endpoint

Students can identify the constraint inputs and see that an interior minimum-KL solution has $h = q$, $T = f$, $\eta = -\lambda$, and $A = \log Z$.

Entry / assumed context

Students know expectation, entropy, KL, and the common exponential-family form. Section 4 estimated parameters in a chosen family; this section instead starts from feature constraints and chooses the distribution.

Narrative arc

Known feature expectations ⟶ closest constrained distribution to q ⟶ uniform reference and entropy ⟶ multiplier proof sketch ⟶ common form.

Visual and notation anchors

- Finite support $\mathcal{X}$ with M states; $q(x) > 0$.
- $E_p[f_k(X)] = F_k$, $k = 1, \ldots, K$.
- $\mathrm{KL}(p\|q) = \sum_{x\in\mathcal{X}} p(x) \log\left(\frac{p(x)}{q(x)}\right)$.
- $T(x) = f(x)$, $h(x) = q(x)$, $\eta = -\lambda$.

Summary design brief

Recombine constraints, the KL criterion, and the resulting exponential form. Use short takeaways for the roles of features, multipliers, and reference. Do not repeat the Lagrangian or add a new Gaussian example.

Continuity / entry context

Open with a new model-construction question after Section 4′s estimation problem. Entropy/KL are review; this is not an inference from MLE concavity. End with the completed construction, without a closing bridge.

Constraints / open pressure

Five slides in 8–10 minutes. Finite discrete support keeps the live proof to ordinary derivatives; assume feasible constraints admitting an interior optimum. Uniform q is proper on this support. Boundary constraints can require limits; no functional calculus or convex duality is taught. Source: Murphy Book 2 §2.4.7, printed pp.43–44, equations (2.251)–(2.256). Follow Murphy's maximization of negative KL, giving $\eta = -\lambda$.

Section 5 — title

Moment constraints and maximum entropy lead to exponential families

Teaching move

Frame the construction question that begins from known feature expectations.

Layout sketch

section-title

footer-title

Content objects

section-title — Section title and subtitle — card

Moment constraints and maximum entropy lead to exponential families

Choose a distribution from expected features and a reference measure.

footer-title — Abbreviated footer title — card

Maximum entropy.

Reveal sketch

Static: section-title, footer-title

Design constraints

Use the established section-title hierarchy and abbreviated footer metadata.

Moment constraints state what we know about an otherwise unknown distribution

Teaching move

Define the supplied feature expectations and feasible distributions before introducing a selection criterion.

Layout sketch

given-information

feature-constraints

probability-constraints

takeaway

Content objects

given-information — Inputs to the construction card

Choose a distribution p on a fixed finite set $\mathcal{X}$. Given: feature functions $f_1, \ldots, f_K$ and expected values $F_1, \ldots, F_K$.

feature-constraints — The information p must preserve card

$$E_p[f_k(X)] = \sum_{x \in \mathcal{X}} p(x) f_k(x) = F_k, \quad k = 1, \ldots, K$$

These values are supplied constraints on the distribution.

probability-constraints — p must also be a distribution card

$$p(x) \geq 0, \quad \sum_{x \in \mathcal{X}} p(x) = 1$$

takeaway — Key points key-points

Feature expectations constrain the unknown distribution but may not determine it.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	given-information	—	Initial meaningful context
STATE 2	+ feature-constraints	entry emphasis	State precisely what is assumed known.
STATE 3	+ probability-constraints	—	Specify feasibility as probabilities as well as moments.
STATE 4	+ takeaway	—	Explain why a selection criterion is needed.

Summary candidates

VISUAL feature-constraints — The starting information of the construction.

Design constraints

F denotes supplied expectations, not automatically the sample statistic from Section 4. Explain that multiple p may satisfy the same constraints; no numerical example or new dataset is needed.

Maximum entropy chooses the least committed distribution satisfying those constraints

Teaching move

State minimum KL to a positive reference and show that uniform reference reduces it to maximizing entropy under the same constraints.

Layout sketch

constraints-recall

minimum-kl

uniform-reference

takeaway

Content objects

constraints-recall — Carried forward recall-card

$$E_p[f_k(X)] = F_k, \quad \sum_x p(x) = 1, \quad p(x) \geq 0$$

From: Previous slide: supplied feature expectations

minimum-kl — The general selection principle card

Given a reference distribution $q(x) > 0$:

$$p^\star = \arg \min_{p:\text{constraints}} \mathrm{KL}(p\|q)$$

$$\mathrm{KL}(p\|q) = \sum_x p(x) \log\left(\frac{p(x)}{q(x)}\right)$$

uniform-reference — Uniform reference gives maximum entropy bordered-card

$$q(x) = \frac{1}{M}, \quad M = |\mathcal{X}|$$

$$\mathrm{KL}(p\|q) = \log M - H(p), \quad H(p) = -\sum_x p(x) \log p(x)$$

Minimizing KL then maximizes entropy.

takeaway — Key points key-points

- Minimum KL chooses the closest feasible distribution to the reference.
- With a uniform reference, the same criterion is maximum entropy.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	constraints-recall	—	Initial meaningful context
STATE 2	+ minimum-kl	entry emphasis	Define the general meaning of least commitment relative to q.
STATE 3	+ uniform-reference	entry emphasis	Identify the entropy special case through an additive constant.
STATE 4	+ takeaway	—	Retain the reference-dependent selection principle.

Summary candidates

VISUAL minimum-kl — The general criterion behind the exponential form.
KEY POINT (takeaway) Uniform reference turns minimum KL into maximum entropy.

Design constraints

Sum over the fixed finite support. Use the conventional zero-probability limit in entropy/KL. A uniform distribution on all of RR is not proper and is not part of this discrete proof. q is a reference, not a fitted parameter.

Lagrange multipliers turn feature constraints into exponential-family natural parameters

Teaching move

Sketch the finite-dimensional multiplier calculation and identify the features, reference, natural weights, and log partition in its solution.

Layout sketch

lagrangian

stationarity

solution

mapping

takeaway

Content objects

lagrangian — Maximize negative KL with the constraints

card

$$J(p,\lambda) = -\sum_x p(x)\log\left(\frac{p(x)}{q(x)}\right) + \lambda_0\left(1-\sum_x p(x)\right) + \sum_{k=1}^{K}\lambda_k\left(F_k - \sum_x p(x) f_k(x)\right)$$

stationarity — At an interior constrained optimum

card

$$\frac{\partial J}{\partial p(x)} = -1 - \log\left(\frac{p(x)}{q(x)}\right) - \lambda_0 - \sum_k \lambda_k f_k(x) = 0$$

solution — Rearrange and normalize

card

$$p^\star(x) = \frac{q(x)}{Z}\exp\left(-\sum_k \lambda_k f_k(x)\right)$$

$$Z = \sum_x q(x)\exp\left(-\sum_k \lambda_k f_k(x)\right)$$

mapping — Match the four roles

schema-card

$$h(x) = q(x), \quad T(x) = f(x), \quad \eta = -\lambda, \quad A(\eta) = \log Z(\eta)$$

$$p^\star(x) = q(x)\exp(\eta^T f(x) - A(\eta))$$

takeaway — Key points key-points

The constrained solution is an exponential family in the supplied features.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	lagrangian	—	Initial meaningful context
STATE 2	+ stationarity	entry emphasis	Differentiate the probability coordinate in the finite vector p.
STATE 3	+ solution	entry emphasis	Exponentiate the stationarity equation and impose total probability one.
STATE 4	+ mapping	—	Read off the reference, statistic, natural parameter, and normalizer.
STATE 5	+ takeaway	—	State the construction result.

Summary candidates

VISUAL mapping — The completed multiplier-to-family correspondence.
KEY POINT (takeaway) Features become sufficient statistics; multiplier negatives become natural parameters.

Design constraints

Treat p as a finite probability vector. Show the three Lagrangian lines together and each subsequent equation group as one state; no functional derivatives or click-by-click algebra. The sign convention is $\eta = -\lambda$ because the objective is negative KL with constraints written as given minus model expectation. Finite multipliers assume an interior solution.

Section 5 — summary

Moment constraints plus maximum entropy produce an exponential-family model

Teaching move

Consolidate the constraint inputs, selection principle, and exponential-form result with their four-role correspondence.

Layout sketch

summary-visual

summary-takeaways

Content objects

summary-visual — Summary visual

section-summary-visual

$$E_p[f_k(X)] = F_k, \quad p^\star = \arg\min_{p:\text{constraints}} \mathrm{KL}(p\|q)$$

$$p^\star(x) = q(x)\exp(\eta^T f(x) - A(\eta))$$

$$h = q, \quad T = f, \quad \eta = -\lambda, \quad A = \log Z$$

Uniform q: minimum KL is maximum entropy.

summary-takeaways — Summary takeaways

section-summary-takeaways

- Supplied features and expectations define the constraints.
- Minimum KL chooses a feasible distribution relative to q.
- The multiplier solution has exponential-family form.

Reveal sketch

Static: summary-visual, summary-takeaways

Rationale

Select the feature constraints, KL criterion, and completed mapping. The summary omits the Lagrangian while retaining the reference and multiplier signs that make the final form meaningful.

Design constraints

Final retrospective teaching move with the explicit summary label. The finite-support and interior assumptions remain those stated in the section; introduce no empirical example or later-application bridge.

SECTION DESIGN NOTES

Section 6 — Maximum likelihood fits a Gamma model to rainfall data

Purpose

Apply the general MLE rule to a real positive-data, two-parameter model.

Endpoint

Identify Gamma statistics $(\log x, x)$, sample aggregates, and their role in a fitted rainfall model.

Entry / assumed context

Return to Section 4's moment matching after the model-construction view of Section 5.

Narrative arc

Rainfall distribution ⟶ Gamma statistics ⟶ aggregate moments ⟶ fitted-density comparison.

Visual and notation anchors

- Shape $\alpha > 0$, rate $\beta > 0$, rainfall $x > 0$ in inches.
- $T(x) = (\log x, x)$, $\eta = (\alpha - 1, -\beta)$.
- $E[\log X] = \psi(\alpha) - \log \beta$; $E[X] = \frac{\alpha}{\beta}$.

Summary design brief

Return to the rainfall histogram with fitted Gamma overlay and a compact two-statistic reminder.

Continuity / entry context

Reconnect to the estimation rule at entry. Keep the empirical fit and retrospective summary as the endpoint.

Constraints / open pressure

Five slides in 7–9 minutes; currently live. Recording or removal remains a later instructor option, not a current delivery change. No Gamma-property survey or optimizer derivation. Figure assets are delegated below. Sources: MIT 18.443 R Project 2, Le Cam–Neyman southern Illinois summer-storm rainfall, 1960–1964; MIT Lecture 4, pp.9–11, for shape/rate MLE. Raw files are available from the Western University course mirror; local import, independent refit, positivity checks, and plot validation remain outstanding. Do not infer or invent numeric sample aggregates.

Section 6 — title

Maximum likelihood fits a Gamma model to rainfall data

Teaching move

Frame a real-data test of the general MLE rule with a two-parameter positive-data model.

Layout sketch

section-title

footer-title

Content objects

section-title — Section title and subtitle — card

Maximum likelihood fits a Gamma model to rainfall data

Two sufficient statistics connect a positive-data fit to the general learning rule.

footer-title — Abbreviated footer title — card

Gamma rainfall fit.

Reveal sketch

Static: section-title, footer-title

Design constraints

Use the established section-title hierarchy and abbreviated footer metadata.

Rainfall amounts are positive and strongly right-skewed

Teaching move

Establish the empirical shape and support of the rainfall sample through its distribution plot.

Layout sketch

rainfall-data

takeaway

Content objects

rainfall-data — Figure asset brief — rainfall histogram — design-card

Visible figure: density-scaled histogram of the five years of southern Illinois storm averages. Horizontal axis: “Rainfall per storm (inches)”; vertical axis: “Density”. Preserve the long positive tail; no synthetic points.

Caption: Le Cam–Neyman summer-storm rainfall, southern Illinois, 1960–1964. **Visible source note:** Le Cam–Neyman data in Rice, 3rd edition; MIT 18.443 R Project 2. Raw-file access: Western University course mirror.

Provenance / authoring: concatenate `illinois60.txt` through `illinois64.txt` from Western University course handouts, `Data/` links. Verify units, sample size, parsing, and strictly positive observations against the MIT project before plotting. Record any discrepancy. Reuse the same binning and axes in the section summary.

takeaway — Key points — key-points

Positive support and right skew motivate the Gamma model.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	rainfall-data	—	Initial meaningful context
STATE 2	+ takeaway	—	Connect observed shape and support to the model choice.

Summary candidates

VISUAL rainfall-data — The observed distribution anchors the final fit comparison.

Design constraints

The figure dominates; prose and source caption are subordinate. No method-of-moments or numerical fitting discussion.

A Gamma model summarizes the rainfall through `x` and `log x`

Teaching move

Rewrite the shape/rate density only far enough to identify its two data statistics and natural coefficients.

Layout sketch

gamma-density

gamma-mapping

takeaway

Content objects

gamma-density — Gamma shape/rate model — card

$$p(x|\alpha,\beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1}\exp(-\beta x), \quad x > 0$$

$$\log p(x|\alpha,\beta) = (\alpha-1)\log x - \beta x + \alpha\log\beta - \log\Gamma(\alpha)$$

gamma-mapping — Match the common form — schema-card

$$T(x) = \begin{pmatrix}\log x \\ x\end{pmatrix}, \quad \eta = \begin{pmatrix}\alpha-1 \\ -\beta\end{pmatrix}$$

$$h(x) = 1 \ \text{ on } \ x > 0, \quad A(\eta) = \log\Gamma(\alpha) - \alpha\log\beta$$

takeaway — Key points — key-points

The Gamma data statistics are log rainfall and rainfall itself.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	gamma-density	—	Initial meaningful context
STATE 2	+ gamma-mapping	entry emphasis	Identify the data statistics and their coefficients.
STATE 3	+ takeaway	—	Retain the nontrivial log statistic.

Summary candidates

VISUAL gamma-mapping — The two-statistic compression underlying the fit.

Design constraints

Use rate beta throughout; never substitute a scale parameter without inversion. The fixed support is positive rainfall.

The Gamma MLE depends on Σx and Σlog x

Teaching move

Apply the existing learning rule to the two Gamma statistics and acknowledge the resulting digamma equation.

Layout sketch

sample-statistics

moment-equations

fitted-comparison

takeaway

Content objects

sample-statistics — Two aggregates determine the sample likelihood — card

$$S_1 = \sum_{n=1}^{N} \log x_n, \quad S_2 = \sum_{n=1}^{N} x_n$$

$$\ell(\alpha, \beta) = (\alpha - 1)S_1 - \beta S_2 - N(\log \Gamma(\alpha) - \alpha \log \beta)$$

moment-equations — Apply moment matching — card

$$\psi(\hat{\alpha}) - \log \hat{\beta} = \frac{S_1}{N}, \quad \frac{\hat{\alpha}}{\hat{\beta}} = \frac{S_2}{N}$$

$$\psi(\alpha) = \frac{\mathrm{d} \log \Gamma(\alpha)}{\mathrm{d}\alpha}$$

fitted-comparison — Figure asset brief — establish the fitted comparison — design-card

Reuse the rainfall histogram with a fitted Gamma density overlaid, using the same axes and binning. Caption: “Gamma MLE fitted to southern Illinois rainfall”. Label MIT-reported $\hat{\alpha} \approx 0.4408$ (dimensionless) and $\hat{\beta} \approx 1.9648\text{in}^{-1}$, pending independent refit. Visible source note: Le Cam–Neyman/Rice data, MIT 18.443 R Project 2; raw files accessed through the Western University mirror. Import the Western University raw files, independently refit, compare the reported values, and validate the plotted curve before production. With shape below one, evaluate the curve only at $x > 0$.

Retained compact anchor: $\mathcal{D} \to (S_1, S_2) \to \left(\hat{\alpha}, \hat{\beta}\right)$.

takeaway — Key points — key-points

Two sample summaries feed the same moment-matching rule; the shape equation is solved numerically.

Reveal sketch

DESIGN REVEAL — structured lowering unresolved

State 1: show `sample-statistics`. State 2: retain it and add `moment-equations` with transient entry emphasis. State 3: replace the detailed likelihood and moment equations with `fitted-comparison`, retaining only the displayed (S_1, S_2) compression anchor and adding `takeaway`. The histogram and fitted curve remain simultaneous in the final state and handout. This replacement gives the empirical result room without shrinking the earlier mathematics. Authoring owns the exact replacement mechanics.

Summary candidates

VISUAL sample-statistics — The empirical sufficient-statistic reduction.
VISUAL fitted-comparison — The fitted result is established before the summary.

Design constraints

State digamma notation and numerical solution; do not derive its properties or methods. No unverified numeric aggregates appear. The final applied result establishes the fit comparison before retrospective reuse.

Section 6 — summary

A low-dimensional sufficient statistic determines a useful rainfall fit

Teaching move

Return to the observed distribution with its Gamma fit and consolidate the data-to-statistics-to-fit chain.

Layout sketch

summary-visual

summary-takeaways

Content objects

summary-visual — Summary figure asset brief — histogram and Gamma fit — design-card

Reuse the exact `fitted-comparison` histogram, overlay, caption, fitted values, and compression anchor established on the preceding slide. No new fitted result, dataset, or empirical conclusion is introduced here.

summary-takeaways — Summary takeaways — section-summary-takeaways

- Gamma uses $\log x$ and x as its sufficient statistics.
- Their sample sums determine the likelihood and its fitted parameters.
- The fitted density is compared against the observed rainfall distribution.

Reveal sketch

Static: summary-visual, summary-takeaways

Rationale

Select the fitted comparison and two-statistic compression already established on Slide 4. The summary retells that completed estimation chain.

Design constraints

Retrospective closure. The fit comparison is provisional until the delegated data/figure checks pass.

Backpressure

Downstream-owned asset requirement: import and validate the recovered raw files, fit the Gamma model, and build both figures. Current design is complete as an asset specification; no fabricated histogram or refit is supplied. Source fit: MIT R Project 2. Source math: MIT Lecture 4, pp.9–11.

SECTION DESIGN NOTES

Section 7 — Categorical distributions extend the same learning rule to multiple outcomes

Purpose

Extend the Bernoulli learning rule to one distribution over K mutually exclusive outcomes.

Endpoint

Read categorical indicators, softmax logits, sample counts, and normalized-count MLE probabilities.

Entry / assumed context

Use Bernoulli indicators, the common form, and Section 4′s moment-matching rule.

Narrative arc

One-hot data ⟶ indicator statistics ⟶ logit-to-probability map ⟶ count and normalize ⟶ summary.

Visual and notation anchors

- $x \in \{1, \ldots, K\}$; $T_k(x) = \mathrm{I}(x = k)$ for all K categories.
- $p_k = P(X = k) = E[T_k(X)]$; $\sum_k p_k = 1$.
- Full logits: $A(\eta) = \log \sum_k e^{\eta_k}$; $p_k = e^{\eta_k - A(\eta)}$.
- $N_k = \sum_{n=1}^{N} T_k(x_n)$; $\hat{p}_k = \frac{N_k}{N}$.

Summary design brief

Pair indicator means and softmax with counts and normalized MLE probabilities.

Continuity / entry context

Reconnect to Bernoulli at entry. End with one categorical distribution learned from counts, without introducing dependence.

Constraints / open pressure

Six slides in 10–12 minutes. Use full K indicators and logits throughout; this is a redundant representation. Explain that adding a common constant changes no probability. Fixing $\eta_K = 0$ is a reference convention, not a switch to a differently sized statistic vector. Zero counts give boundary fitted probabilities and no finite full-logit representation. Source: Murphy Book 2 §2.4.2.2, printed p.36, equations (2.174)–(2.186). The full-logit form restates (2.174); reference fixing connects to (2.183)–(2.186).

Section 7 — title

Categorical distributions extend the same learning rule to multiple outcomes

Teaching move

Frame the K-outcome extension of Bernoulli and reuse of the general learning rule.

Layout sketch

section-title

footer-title

Content objects

section-title — Section title and subtitle — card

Categorical distributions extend the same learning rule to multiple outcomes

One-hot indicators turn observations into normalized counts.

footer-title — Abbreviated footer title — card

Categorical learning.

Reveal sketch

Static: section-title, footer-title

Design constraints

Use the established section-title hierarchy and abbreviated footer metadata.

Categorical data replace one binary indicator with a one-hot vector

Teaching move

Make mutually exclusive category membership visible as one active coordinate.

Layout sketch

support

one-hot-example

takeaway

Content objects

support — One outcome among K categories — card

$$x \in \{1, \ldots, K\}, \quad T_k(x) = \mathrm{I}(x = k)$$

one-hot-example — Course-constructed three-category illustration — schema-card

$$x = 1 \to \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad x = 2 \to \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad x = 3 \to \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\sum_{k=1}^{K} T_k(x) = 1$$

takeaway — Key points — key-points

Each observation activates exactly one category indicator.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	support	—	Initial meaningful context
STATE 2	+ one-hot-example	—	Compare the three possible encoded observations simultaneously.
STATE 3	+ takeaway	—	State the mutual-exclusion structure.

Summary candidates

VISUAL one-hot-example — Concrete meaning of the indicator statistic.

Design constraints

The illustration is constructed for the course, not a Murphy example. Keep all three vectors simultaneous; no distribution fitting yet.

Category indicators are the sufficient statistics

Teaching move

Show the categorical density in indicator form and identify indicator expectations as probabilities.

Layout sketch

categorical-density

indicator-means

takeaway

Content objects

categorical-density — The selected category determines the probability

card

$$p(x|\boldsymbol{p}) = \prod_{k=1}^{K} p_k^{T_k(x)}, \quad h(x) = 1 \quad \text{on} \quad \{1, ..., K\}$$

$$T(x) = \begin{pmatrix} T_1(x) \\ ... \\ T_K(x) \end{pmatrix}$$

indicator-means — The model moments are category probabilities

card

$$E[T_k(X)] = P(X = k) = p_k, \quad k = 1, ..., K$$

$$E[T(X)] = \boldsymbol{p}$$

takeaway — Key points

key-points

The categorical mean statistic is the vector of category probabilities.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	categorical-density	—	Initial meaningful context
STATE 2	+ indicator-means	entry emphasis	Connect each indicator expectation to its category probability.
STATE 3	+ takeaway	—	Read the entire mean vector.

Summary candidates

VISUAL indicator-means — Connects statistics directly to the eventual estimator.

Design constraints

Keep the full K-vector convention. All probabilities are positive for the finite-logit mapping developed next.

Natural parameters become category probabilities through softmax

Teaching move

Normalize category log weights and explain their common-shift redundancy.

Layout sketch

logit-form

softmax

reference-convention

takeaway

Content objects

logit-form — Full-logit exponential-family form

card

$$p(x|\eta) = \exp\left(\sum_{k=1}^{K} \eta_k T_k(x) - A(\eta)\right)$$

$$A(\eta) = \log \sum_{j=1}^{K} e^{\eta_j}$$

softmax — Normalized log weights

schema-card

$$p_k = \frac{e^{\eta_k}}{\sum_{j=1}^{K} e^{\eta_j}}, \quad \boldsymbol{p} = \operatorname{softmax}(\eta)$$

$$\operatorname{softmax}(\eta + c\mathbf{1}) = \operatorname{softmax}(\eta)$$

reference-convention — One reference fixes the shared shift

card

$$\eta_K = 0 \to \eta_k = \log\left(\frac{p_k}{p_K}\right), \quad k < K$$

$$A(\eta) = \log\left(1 + \sum_{j=1}^{K-1} e^{\eta_j}\right)$$

takeaway — Key points

key-points

Softmax turns natural log weights into probabilities; a common shift changes none of them.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	logit-form	—	Initial meaningful context
STATE 2	+ softmax	entry emphasis	Show the probability map and its shared-shift invariance.
STATE 3	+ reference-convention	—	Explain how a reference category selects one representative.
STATE 4	+ takeaway	—	Retain the map and redundancy.

Summary candidates

VISUAL softmax — The categorical natural-to-mean map.

Design constraints

Keep full T and eta vectors, with etaK fixed only in the reference convention. Do not silently use a K−1 statistic vector with a K-term log partition.

Category counts are sufficient statistics, and normalization gives their MLE

Teaching move

Aggregate indicators into category counts and apply moment matching in the same teaching move.

Layout sketch

category-counts

matched-probabilities

boundary-note

takeaway

Content objects

category-counts — Aggregate the indicators card

$$N_k = \sum_{n=1}^{N} \mathrm{I}(x_n = k), \quad T(\mathcal{D}) = \begin{pmatrix} N_1 \\ \dots \\ N_K \end{pmatrix}$$

$$\sum_{k=1}^{K} N_k = N > 0$$

matched-probabilities — Match expected indicators to observed fractions card

$$E_{\hat{p}}[T_k(X)] = \hat{p}_k = \frac{N_k}{N}$$

boundary-note — Zero counts are boundary fits card

If $N_k = 0$, then $\hat{p}_k = 0$; finite logits cannot represent that boundary.

takeaway — Key points key-points

Categorical MLE is count and normalize, reusing the general moment-matching rule.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	category-counts	—	Initial meaningful context
STATE 2	+ matched-probabilities	entry emphasis	Apply the indicator-mean identity to the sample counts.
STATE 3	+ boundary-note, takeaway	—	Retain the estimator, including zero counts.

Summary candidates

VISUAL matched-probabilities — The categorical learning result.

Design constraints

No fresh Lagrange derivation or smoothing. Distinguish probability MLE from existence of a finite natural-coordinate MLE.

Section 7 — summary

For categorical models, learning means count and normalize

Teaching move

Consolidate categorical statistics, probabilities, natural coordinates, and normalized-count learning.

Layout sketch

summary-visual

summary-takeaways

Content objects

summary-visual — Summary visual

section-summary-visual

$$T_k(x) = \mathrm{I}(x = k), \quad E[T_k(X)] = p_k$$

$$p_k = \frac{e^{\eta_k}}{\sum_j e^{\eta_j}}$$

$$N_k = \sum_{n=1}^{N} T_k(x_n) \to \hat{p}_k = \frac{N_k}{N}$$

summary-takeaways — Summary takeaways

section-summary-takeaways

- One-hot indicators encode mutually exclusive outcomes.
- Softmax maps log weights to category probabilities.
- Normalized category counts give the probability MLE.

Reveal sketch

Static: summary-visual, summary-takeaways

Rationale

Select indicator means, softmax, and the normalized-count estimator. Omit the three-category illustration and reference-fixing details from the compact summary.

Design constraints

Final retrospective teaching move, explicitly labeled as a summary. No conditional sequence model or receiving-section teaser.

SECTION DESIGN NOTES

Section 8 — A Markov model learns one categorical distribution for each preceding state

Purpose

Learn a first-order sequence model by reusing categorical estimation for each conditioning state.

Endpoint

Read the factorization, transition rows, transition counts, row MLE, sequence scoring, and repeated conditional sampling.

Entry / assumed context

Section 7 established one categorical distribution learned from counts. Here the preceding state selects which categorical row applies.

Narrative arc

Order matters → previous-state context → categorical rows → transition counts → normalized-row MLE → text counts and fitted rows → scoring and sampling → retrospective summary.

Visual and notation anchors

- $A_{jk} = P(X_t = k | X_{t-1} = j)$; row j is preceding, column k is next.
- $\pi_j = P(X_1 = j)$; $p(x_{1:T}) = \pi_{x_1} \prod_{t=2}^{T} A_{x_{t-1}, x_t}$.
- $N_{jk} = \sum_{t=2}^{T} \mathrm{I}(x_{t-1} = j, x_t = k)$; $N_j = \sum_k N_{jk}$.
- $\hat{A}_{jk} = \frac{N_{jk}}{N_j}$ when $N_j > 0$.

Summary design brief

Select the count-to-row-normalization chain, then recall scoring and sampling as two uses of the same learned rows. No notebook task belongs inside the summary.

Continuity / entry context

Open with the limitation of independent categorical observations for ordered data. End with what the fitted first-order model does; the notebook handoff remains a separate lecture-level activity.

Constraints / open pressure

Eleven slides in 20–24 minutes. Finite state space and time-homogeneous transitions. Transition counts suffice for A; initial-distribution estimation needs separate start-state counts. Unobserved predecessor rows are not identified; zero transition factors give zero unsmoothed sequence probability. No stationary distributions, matrix powers, or live smoothing extension. Sources: Murphy Book 2 §§2.6.1–2.6.3, printed pp.47–50, equations (2.265), (2.274)–(2.278), Fig.2.15(a), and Fig.2.17.

Approved sampling source: course-generated first-order Time Machine output from the validated matrix.

Section 8 — title

A Markov model learns one categorical distribution for each preceding state

Teaching move

Frame a sequence model as a collection of conditional categorical rows.

Layout sketch

section-title

footer-title

Content objects

section-title — Section title and subtitle — card

A Markov model learns one categorical distribution for each preceding state

Transition counts fit conditional categorical models for sequence data.

footer-title — Abbreviated footer title — card

Markov learning.

Reveal sketch

Static: section-title, footer-title

Design constraints

Use the established section-title hierarchy and abbreviated footer metadata.

Sequence models need conditional probabilities, not independent draws

Teaching move

Show that identical category counts can conceal different order and transition structure.

Layout sketch

same-counts | **different-order**

takeaway

Content objects

same-counts — Same categorical totals schema-card

Course-constructed sequences: `ab` and `ba`. Both contain one a and one b.

$$p_{\text{IID}}(\text{ab}) = p_a p_b = p_{\text{IID}}(\text{ba})$$

different-order — Different conditional transitions schema-card

`ab` contains $a \to b$; `ba` contains $b \to a$.

$$p(\text{ab}) = \pi_a A_{ab}, \quad p(\text{ba}) = \pi_b A_{ba}$$

Ordered models can distinguish the two sequences.

takeaway — Key points key-points

Category totals describe frequencies; conditional transitions describe order.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	same-counts	—	Initial meaningful context
STATE 2	+ different-order	—	Compare the two order-sensitive transition factors.
STATE 3	+ takeaway	—	State what a sequence model must retain.

Design constraints

The toy comparison is course-constructed. Explain pi as an initial probability and A as a conditional transition; their general definitions follow immediately. No empirical text probabilities are invented.

A first-order Markov model keeps only the previous state as context

Teaching move

State the first-order conditional assumption and factorization, supported by a state-transition schema.

Layout sketch

markov-assumption

factorization

state-diagram

takeaway

Content objects

markov-assumption — First-order context — card

$$p(x_t|x_{1:t-1}) = p(x_t|x_{t-1})$$

factorization — Joint sequence probability — card

$$p(x_{1:T}) = p(x_1)\prod_{t=2}^{T} p(x_t|x_{t-1})$$

$$\pi_j = P(X_1 = j)$$

state-diagram — Schema asset brief — two-state chain — design-card

Adapt Murphy Fig.2.15(a): states 1 and 2; arrows 1⟶2 labeled A12, 2⟶1 labeled A21, and self-loops labeled A11 and A22. These correspond to Murphy's alpha, beta, 1–alpha, 1–beta respectively. All four legal transitions remain visible. Caption: "A state chooses the next-state distribution". Draw as a restrained native course schema; no numeric weights are assigned.

takeaway — Key points — key-points

The model uses only the preceding state to predict the next state.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	markov-assumption	—	Initial meaningful context
STATE 2	+ factorization	entry emphasis	Apply the conditional assumption to the chain rule.
STATE 3	+ state-diagram	—	Make the transition structure concrete.
STATE 4	+ takeaway	—	Retain the first-order assumption.

Summary candidates

VISUAL factorization — The model's probability structure.

Design constraints

Distinguish a state-transition diagram from a time-indexed dependency graph. T denotes sequence length here. No graphical conditional-independence claims beyond the stated Markov assumption.

Each transition-matrix row is one categorical distribution

Teaching move

Interpret rows as next-state categorical probabilities for a fixed preceding state.

Layout sketch

transition-definition

row-distribution

takeaway

Content objects

transition-definition — A time-homogeneous K-state model — card

$$A_{jk} = P(X_t = k | X_{t-1} = j), \quad A \in \mathbb{R}^{K \times K}$$

row-distribution — Hold the preceding state j fixed — schema-card

$$A_{j\cdot} = (A_{j1}, \ldots, A_{jK}), \quad A_{jk} \geq 0, \quad \sum_{k=1}^{K} A_{jk} = 1$$

$$X_t \mid (X_{t-1} = j) \sim \mathrm{Cat}(A_{j\cdot})$$

Rows: preceding state. Columns: next state.

takeaway — Key points — key-points

A Markov chain has one categorical next-state distribution for each preceding state.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	transition-definition	—	Initial meaningful context
STATE 2	+ row-distribution	entry emphasis	Read a fixed row as the familiar categorical model.
STATE 3	+ takeaway	—	State the reuse of Section 7.

Summary candidates

VISUAL row-distribution — The conditional categorical interpretation.

Design constraints

A is now the transition matrix, not the earlier log-partition function A(eta). Tie the same row across time; do not introduce stationarity or a stationary distribution.

Transition counts are the sufficient statistics for a Markov chain

Teaching move

Count observed transitions and group the transition likelihood into row-wise categorical factors.

Layout sketch

transition-counts

grouped-likelihood

initial-note

takeaway

Content objects

transition-counts — Count j⟶k transitions in one observed sequence — card

$$N_{jk} = \sum_{t=2}^{T} \mathrm{I}(x_{t-1} = j, x_t = k), \quad N_j = \sum_k N_{jk}$$

grouped-likelihood — Collect identical conditional factors — card

$$p(x_{2:T}|x_1, A) = \prod_{t=2}^{T} A_{x_{t-1}, x_t} = \prod_{j,k} A_{jk}^{N_{jk}}$$

$$\ell(A) = \sum_{j,k} N_{jk} \log A_{jk}$$

initial-note — Initial probabilities are a separate part — card

Transition counts suffice for A. Estimating pi across separate sequences uses their initial-state counts, not these transition counts.

takeaway — Key points — key-points

The transition likelihood depends on the sequence through the j⟶k counts.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	transition-counts	—	Initial meaningful context
STATE 2	+ grouped-likelihood	entry emphasis	Group repeated factors by preceding and next state.
STATE 3	+ initial-note, takeaway	—	State exactly which model parameters the counts inform.

Summary candidates

VISUAL transition-counts — The observed statistics used to learn A.

Design constraints

Conditional likelihood holds the initial state fixed. For a collection of independent sequences, sum these counts within each sequence; never count a transition across a sequence boundary.

Normalizing each row gives the maximum-likelihood transition matrix

Teaching move

Apply categorical count normalization independently to each observed predecessor row.

Layout sketch

categorical-recall

row-mle

unobserved-row

takeaway

Content objects

categorical-recall — Carried forward recall-card

$$\hat{p}_k = \frac{N_k}{N}$$

From: Section 7: categorical MLE

row-mle — One row, one count-and-normalize fit card

$$\hat{A}_{jk} = \frac{N_{jk}}{N_j}, \quad N_j > 0$$

$$\sum_k \hat{A}_{jk} = 1$$

unobserved-row — What the data do not determine card

If $N_j = 0$, row j is unidentified. If $N_j > 0$ but $N_{jk} = 0$, the unsmoothed MLE gives $\hat{A}_{jk} = 0$.

takeaway — Key points key-points

Normalize observed transition counts within each preceding-state row.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	categorical-recall	—	Initial meaningful context
STATE 2	+ row-mle	entry emphasis	Reuse count normalization for the conditional distribution.
STATE 3	+ unobserved-row, takeaway	—	Retain the learning rule and its data-coverage condition.

Summary candidates

VISUAL row-mle — The transition learning rule.

Design constraints

No fresh Lagrange derivation or silent smoothing. Keep Nj>0 adjacent to the estimator.

Character bigrams turn text into a transition-count matrix

Teaching move

Make the corpus-to-counts operation concrete through the empirical bigram matrix.

Layout sketch

text-to-counts

bigram-counts

takeaway

Content objects

text-to-counts — Count adjacent characters schema-card

$$(x_{t-1}, x_t) = (j, k) \rightarrow N_{jk} \quad \text{increases by } 1$$

bigram-counts — Figure asset brief — Time Machine raw counts design-card

Dominant empirical count matrix corresponding to Murphy Fig.2.17(a), trained on H. G. Wells's The Time Machine. Character order follows decreasing unigram frequency, with space explicitly labeled "space". Axes: "Preceding character (row)" and "Next character (column)". Highlight the verified source example e⟶space, the most frequent bigram. Use a clearly labeled count scale and verify any numeric annotations. Visible source note: Murphy Book 2 Fig.2.17(a), The Time Machine bigrams; course reconstruction if regenerated.

Authoring: obtain the source corpus and inspect the cited `bigram_hinton_diagram.ipynb` preprocessing/count construction. Record corpus version, preprocessing, alphabet, and boundary handling. Regenerate rather than infer numerical counts from plotted square areas. Use the same verified counts, order, and labels on the normalized-matrix slide.

takeaway — Key points key-points

Each matrix cell counts how often its next character followed its preceding character.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	text-to-counts	—	Initial meaningful context
STATE 2	+ bigram-counts	—	Inspect the real corpus counts and a verified character transition.
STATE 3	+ takeaway	—	State what a cell represents.

Summary candidates

VISUAL bigram-counts — The empirical learning input.

Design constraints

No unverified cell values or replacement toy matrix is presented as Murphy's empirical data. The figure is a delegated asset.

Row normalization turns bigram counts into next-character distributions

Teaching move

Show the fitted conditional probabilities produced by normalizing the same empirical count matrix.

Layout sketch

row-rule

bigram-probabilities

takeaway

Content objects

row-rule — Carried forward recall-card

$$\hat{A}_{jk} = \frac{N_{jk}}{\sum_l N_{jl}}, \quad N_j > 0$$

From: Earlier in this section: row MLE

bigram-probabilities — Figure asset brief — validated normalized bigrams design-card

Use the same counts, alphabet order, and axes as `bigram-counts`; display conditional probabilities with an explicit probability scale. Highlight one preceding-character row and mark its row sum as 1 after validation. Caption: "Next-character distributions from row-normalized bigram counts". Source note: "Course reconstruction of Murphy Fig.2.17(b) from the verified The Time Machine counts".

Normalize using `row_sums = counts.sum(axis=1, keepdims=True)` and divide each observed row by its own total. Verify nonnegativity and unit sums for every observed row; mark unobserved rows as unidentified. The source notebook's broadcasting expression is not a valid implementation to copy. Do not assume its normalized plot passed these checks.

takeaway — Key points key-points

A row is a fitted distribution over the next character, conditional on its predecessor.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	row-rule	—	Initial meaningful context
STATE 2	+ bigram-probabilities	—	Inspect the empirical counterpart of categorical row fitting.
STATE 3	+ takeaway	—	Read one row as the learned conditional model.

Summary candidates

VISUAL bigram-probabilities — The empirical fitted transition model.

Design constraints

Keep the preceding/next orientation explicit. Correct row normalization is an authoring-owned data-validation requirement, not a change to the approved teaching move.

The fitted Markov model assigns probabilities to new sequences

Teaching move

Reuse the learned transition factors to score a new sequence.

Layout sketch

sequence-score

score-example

zero-factor

takeaway

Content objects

sequence-score — Multiply the selected transition probabilities — card

$$p\left(x_{1:T}|\pi, \hat{A}\right) = \pi_{x_1} \prod_{t=2}^{T} \hat{A}_{x_{t-1}, x_t}$$

$$\log p\left(x_{1:T}|\pi, \hat{A}\right) = \log \pi_{x_1} + \sum_{t=2}^{T} \log \hat{A}_{x_{t-1}, x_t}$$

score-example — Course-constructed symbolic score — schema-card

$$p\left(\text{aba}|\pi, \hat{A}\right) = \pi_a \hat{A}_{ab} \hat{A}_{ba}$$

zero-factor — Unseen transitions under an unsmoothed fit — card

A zero initial or transition factor makes the sequence probability zero.

takeaway — Key points — key-points

The same fitted rows assign a probability to any sequence whose required rows are defined.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	sequence-score	—	Initial meaningful context
STATE 2	+ score-example	entry emphasis	Read the factors for a short symbolic sequence.
STATE 3	+ zero-factor, takeaway	—	State what scoring means, including unsmoothed zero factors.

Summary candidates

VISUAL sequence-score — Scoring is one use of the fitted model.

Design constraints

pi is a separately declared initial distribution; transition counts do not silently fit it. Numeric scores require validated pi and all required rows. Do not add smoothing or likelihood-comparison exercises.

Sampling the fitted transitions generates new text

Teaching move

Show repeated categorical row sampling and its generated first-order text output.

Layout sketch

sampling-rule

generated-text

takeaway

Content objects

sampling-rule — Repeated conditional sampling — schema-card

$$X_1 \sim \mathrm{Cat}(\pi), \quad (X_t | X_{t-1} = j) \sim \mathrm{Cat}\left(\hat{A}_{j\cdot}\right)$$

Current character j ⟶ sample from row j ⟶ new current character.

generated-text — Asset brief — course-generated first-order sample — design-card

Generate a short fixed-seed sample from the validated Time Machine first-order character transition matrix used above, with a documented initial distribution or given starting character. Preserve spaces. Visible caption: “Course-generated sample: first-order character model, trained on The Time Machine”. Show the actual unedited sample, not invented English or selected prose presented as model output. Keep its short-memory character visible; do not promise coherent language.

Source distinction: Murphy Fig.2.16 is a 10-gram model trained on the King James Bible. Its printed excerpt is not output from this first-order fit. Authoring verifies the seed, model parameters, starting condition, and defined rows before producing this asset.

takeaway — Key points — key-points

Generation repeatedly samples the fitted categorical row selected by the current character.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	sampling-rule	—	Initial meaningful context
STATE 2	+ generated-text	—	Inspect actual output of the stated first-order fitted model.
STATE 3	+ takeaway	—	Connect the generated sequence to repeated row sampling.

Summary candidates

VISUAL sampling-rule — Generation is the second use of the same fitted rows.

Section 8 — summary

From text to a fitted Markov model

Teaching move

Consolidate the text-to-counts-to-rows learning chain and the two uses of the learned model.

Layout sketch

summary-visual

summary-takeaways

Content objects

summary-visual — Summary visual — section-summary-visual

Text ⟶ transition counts ⟶ normalized rows ⟶ scoring / sampling.

$$N_{jk} \to \hat{A}_{jk} = \frac{N_{jk}}{N_j}, \quad N_j > 0$$

$$p\left(x_{1:T} | \pi, \hat{A}\right) = \pi_{x_1} \prod_{t=2}^{T} \hat{A}_{x_{t-1}, x_t}$$

$$(X_t | X_{t-1} = j) \sim \mathrm{Cat}\left(\hat{A}_{j\cdot}\right)$$

summary-takeaways — Summary takeaways — section-summary-takeaways

- Each preceding state selects one categorical next-state distribution.
- Transition counts learn those rows by count and normalize.
- The fitted rows support sequence probabilities and conditional sampling.

Reveal sketch

Static: summary-visual, summary-takeaways

Rationale

Select the row MLE, factorization, and sampling rule. The established empirical matrices are supporting evidence; omit large repeated figures and generated excerpts from this compact summary.

Design constraints

Final retrospective section teaching move. No new claim or exercise is introduced. The separate lecture-level notebook handoff in slide_organization.md follows outside this slide and outside the section contract.